# (FP2) 5. Complex numbers 作业

NO. 1

Find all the roots of the equation $(w+1)^6 = 1$, giving your answers in the form $x+iy$ where $x$ and $y$ are real and exact. [4]

A level FURTHER MATHEMATICS 2020/Winter/Paper2_2/Q3

Mark scheme

| Question | Answer | Marks | Guidance |
|---|---|---|---|
| 3 | $w+1=\cos\frac{2n}{6}\pi+i\sin\frac{2n}{6}\pi, \quad n=0,\ldots,5$ | **M1 A1** | Uses the 6th roots of unity. |
| | $w=0,-\frac{1}{2}\pm\frac{1}{2}i\sqrt{3}, \ -\frac{3}{2}\pm\frac{1}{2}i\sqrt{3},-2$ | **A1 A1** | A1 for 3 correct roots. A1 Gives all six roots of the equation in exact form. |
| | | 4 | |

Explanation

3.

Let $w = e^{i\theta} - 1, 0 \le \theta < 2\pi$

$e^{6i\theta} = 1$

$cos\,6\theta + i\,sin\,6\theta = 1$

$cos\,6\theta = 1\ and\ sin6\theta = 0$

$0 \le 6\theta < 2\pi$

$6\theta = 0, 2\pi, 4\pi, 6\pi, 8\pi, 10\pi$

$\theta = 0, \frac{\pi}{3}, \frac{2\pi}{3}, \pi, \frac{4\pi}{3}, \frac{5\pi}{3}$ **M1 A1**

$w = e^{i\theta} - 1 = cos\,\theta + i\,sin\theta - 1$

$= 0, -\frac{1}{2} + i\frac{\sqrt{3}}{2}, -\frac{3}{2} + i\frac{\sqrt{3}}{2}, -2, -\frac{3}{2} - i\frac{\sqrt{3}}{2}, -\frac{1}{2} - i\frac{\sqrt{3}}{2}$ **A1 A1**

Comment

Almost all candidates wrote down the sixth roots of unity and most candidates adjusted these accurately to give all six solutions.

NO. 2

a. Use de Moivre's theorem to show that $\sin^6\theta = -\frac{1}{32}(\cos 6\theta - 6\cos 4\theta + 15\cos 2\theta - 10)$. [6]

It is given that $\cos^6\theta = \frac{1}{32}(\cos 6\theta + 6\cos 4\theta + 15\cos 2\theta + 10)$.

b. Find the exact value of $\int_0^{\frac{1}{3}\pi} \left(\cos^6(\frac{1}{4}x) + \sin^6(\frac{1}{4}x)\right) \mathrm{d}x$. [4]

c. Express each root of the equation $16c^6 + 16(1 - c^2)^3 - 13 = 0$ in the form $\cos k\pi$, where $k$ is a rational number. [5]

A level FURTHER MATHEMATICS 2020/Summer/Paper2_3/Q8

Mark scheme

| Question | Answer | Marks |
|---|---|---|
| a | $z - z^{-1} = 2\mathrm{i}\sin\theta$ | **B1** |
| | $(z - z^{-1})^6 = (z^6 + z^{-6}) - 6(z^4 + z^{-4}) + 15(z^2 + z^{-2}) - 20$ | **M1 A1** |
| | $(2\mathrm{i}\sin\theta)^6 = 2\cos 6\theta - 6(2\cos 4\theta) + 15(2\cos 2\theta) - 20$ | **M1 A1** |
| | $\sin^6\theta = \frac{1}{32}(-\cos 6\theta + 6\cos 4\theta - 15\cos 2\theta + 10)$ | **A1** |
| | | **6** |
| b | $\int_0^{\frac{1}{3}\pi} \cos^6\frac{1}{4}x + \sin^6\frac{1}{4}x\,\mathrm{d}x = \frac{1}{8}\int_0^{\frac{1}{3}\pi} 3\cos x + 5\,\mathrm{d}x$ | **M1 A1** |
| | $\frac{1}{8}\left[3\sin x + 5x\right]_0^{\frac{1}{3}\pi} = \frac{1}{8}\left(\frac{5}{3}\pi + \frac{3}{2}\sqrt{3}\right).$ | **M1 A1** |
| | | **4** |
| c | $c = \cos\theta \Rightarrow 1 - c^2 = \sin^2\theta$ | **B1** |
| | $16c^6 + 16(1-c^2)^3 - 13 = 0 \Rightarrow 6\cos 4\theta - 3 = 0$ | **M1 A1** |
| | $4\theta = \frac{1}{3}\pi, \frac{5}{3}\pi, \frac{7}{3}\pi, \frac{11}{3}\pi$ | **M1** |
| | $c = \cos\left(\frac{1}{12}\pi\right), \cos\left(\frac{5}{12}\pi\right), \cos\left(\frac{7}{12}\pi\right), \cos\left(\frac{11}{12}\pi\right)$ | **A1** |
| | | 5 |

Explanation

a.

let $z = e^{i\theta} = \cos\theta + i\sin\theta$

$z + \frac{1}{z} = z + z^{-1} = \cos\theta + i\sin\theta + \cos(-\theta) + i\sin(-\theta)$

$= \cos\theta + i\sin\theta + \cos\theta - i\sin\theta = 2\cos\theta$

Similarly, $z - \frac{1}{z} = \cos\theta + i\sin\theta - \cos\theta + i\sin\theta = 2i\sin\theta$, **B1**

$z^2 + \frac{1}{z^2} = \cos 2\theta + i\sin 2\theta + \cos 2\theta - i\sin 2\theta = 2\cos 2\theta$, and etc.

$\left(z - \frac{1}{z}\right)^6 = z^6 - 6C_1 z^5 \cdot \frac{1}{z} + 6C_2 z^4 \cdot \frac{1}{z^2} - 6C_3 z^3 \cdot \frac{1}{z^3} + 6C_4 z^2 \cdot \frac{1}{z^4} -$
$6C_5 z^1 \cdot \frac{1}{z^5} + z^{-6}$

$= z^6 + z^{-6} - 6(z^4 + z^{-4}) + 15(z^2 + z^{-2}) - 20$ **M1 A1**

$= 2(cos6\theta - 6\cos 4\theta + 15\cos 2\theta - 20)$

So $(2i)^6 sin^6\theta = 2(cos6\theta - 6\cos 4\theta + 15\cos 2\theta) - 20$ **M1 A1**

$-64\, sin^6\theta = 2(cos6\theta - 6\cos 4\theta + 15\cos 2\theta) - 20$

$sin^6\theta = -\frac{1}{32}(cos6\theta - 6\cos 4\theta + 15\cos 2\theta - 10)$ **A1**

b)

compare the identities of $sin^6\theta$ and $cos^6\theta$

$cos^6\left(\frac{1}{4}x\right) + sin^6\left(\frac{1}{4}x\right) = \frac{1}{32}\left(2 \times 6\, cos\, 4\left(\frac{1}{4}x\right) + 2 \times 10\right)$

$= \frac{1}{8}(3\cos x + 5)$ **M1 A1**

$\int_0^{\frac{\pi}{3}} \left(cos^6\left(\frac{1}{4}x\right) + sin^6\left(\frac{1}{4}x\right)\right) dx = \frac{1}{8}\int_0^{\frac{\pi}{3}} (3\cos x + 5)\, dx = \frac{1}{8}[3\sin x +$ **M1 A1**
$5x]_0^{\frac{\pi}{3}} = \frac{1}{8}\left(3 \cdot \frac{\sqrt{3}}{2} + \frac{5\pi}{3}\right) = \frac{3\sqrt{3}}{16} + \frac{5\pi}{24}$

c)

try $c = cos\,\theta,\ 0 \le \theta \le \pi$

$16\, cos^6\theta + 16(1 - cos^2\theta)^3 - 13 = 0$

$16\, cos^6\theta + 16 sin^6\theta - 13 = 0$ **B1**

$16 \times \frac{1}{8}(3\, cos\, 4\theta + 5) - 13 = 0$

$6\cos 4\theta = 3$ **M1 A1**

$\cos 4\theta = \frac{1}{2}$

$0 \leq 4\theta \leq 4\pi$

$4\theta = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}, \frac{11\pi}{3}$ **M1**

$\theta = \frac{\pi}{12}, \frac{5\pi}{12}, \frac{7\pi}{12}, \frac{11\pi}{12}$

$c = \cos\left(\frac{\pi}{12}\right), \cos\left(\frac{5\pi}{12}\right), \cos\left(\frac{7\pi}{12}\right), \cos\left(\frac{11\pi}{12}\right)$ **A1**

(This is a quartic equation, not a hexic equation)

NO. 3

a. Find the roots of the equation $z^3 = -1 - i$, giving your answers in the form $re^{i\theta}$, where $r > 0$ and $0 \leq \theta \leq 2\pi$. [5]

Let $w = z_1^{3k} + z_2^{3k} + z_3^{3k}$, where $k$ is a positive integer and $z_1$, $z_2$, $z_3$ are the roots of $z^3 = -1 - i$.

b. Express $w$ in the form $Re^{i\alpha}$, where $R > 0$, giving $R$ and $\alpha$ in terms of $k$. [3]

A level FURTHER MATHEMATICS 2020/Summer/Paper2_1/Q3

Mark scheme

| Question | Answer | Marks |
|---|---|---|
| a | $z^3 = -1 - i = 2^{\frac{1}{2}} e^{i\frac{5}{4}\pi}$ | B1 |
| | $z_1 = 2^{\frac{1}{6}} e^{i\frac{5}{12}\pi}$ | M1 A1 |
| | $z_2 = 2^{\frac{1}{6}} e^{i\frac{13}{12}\pi}$, $z_3 = 2^{\frac{1}{6}} e^{i\frac{21}{12}\pi}$ | A1 A1 |
| | | 5 |
| b | $z_1^{3k} + z_2^{3k} + z_3^{3k} = 3\left(2^{\frac{1}{2}k} e^{i\frac{5}{4}k\pi}\right)$ | M1 |
| | $R = \lvert w \rvert = 3\left(2^{\frac{1}{2}k}\right)$ | A1 |
| | $\alpha = \frac{5}{4}k\pi$ | A1 |
| | | 3 |

Explanation

a.

$-1 - i = \sqrt{2}\left(cos\left(\frac{5}{4}\pi\right) + i\,sin\left(\frac{5}{4}\pi\right)\right)$ **B1**

$z = re^{i\theta} = r(cos\theta + i\,sin\theta), \quad 0 \leq \theta < 2\pi$

$z^3 = \left(re^{i\theta}\right)^3 = r^3 e^{i3\theta} = r^3(cos3\theta + i\,sin3\theta)$

So $r^3(cos3\theta + i\,sin3\theta) = \sqrt{2}\left(cos\left(\frac{5}{4}\pi\right) + i\,sin\left(\frac{5}{4}\pi\right)\right)$

$r = 2^{\frac{1}{6}}$

$0 \leq 3\theta < 6\pi$

$3\theta = \frac{5}{4}\pi, \frac{13}{4}\pi, \frac{21}{4}\pi$

$\theta = \frac{5}{12}\pi, \frac{13}{12}\pi, \frac{21}{12}\pi$

$z = 2^{\frac{1}{6}}e^{\frac{5}{12}\pi i}, 2^{\frac{1}{6}}e^{\frac{13}{12}\pi i}2^{\frac{1}{6}}e^{\frac{21}{12}\pi i}$ **M1 A1A1 A1**

b)

$z^3 = -1 - i = \sqrt{2}\left(cos\left(\frac{5}{4}\pi\right) + i\,sin\left(\frac{5}{4}\pi\right)\right) = \sqrt{2}e^{i\frac{5}{4}\pi}$

So $z_1^3 = z_2^3 = z_3^3 = \sqrt{2}\left(cos\left(\frac{5}{4}\pi\right) + i\,sin\left(\frac{5}{4}\pi\right)\right) = \sqrt{2}e^{i\frac{5}{4}\pi}$

$w = 3\left(\sqrt{2}e^{i\frac{5}{4}\pi}\right)^k = 3 \cdot 2^{\frac{k}{2}}e^{i\frac{5}{4}k\pi}$ **M1**

R $= 3 \cdot 2^{\frac{k}{2}}$ and $\alpha = \frac{5}{4}k\pi$ **A1A1**

NO. 4

i. Use de Moivre's theorem to show that

$$\cos 4\theta = \cos^4\theta - 6\cos^2\theta\sin^2\theta + \sin^4\theta. \quad [3]$$

ii. Hence find all the roots of the equation

$$x^4 - 6x^2 + 1 = 0$$

in the form $\tan q\pi$, where $q$ is a positive rational number. [5]

A level FURTHER MATHEMATICS 2018/Summer/Paper1_3/Q3

Mark scheme

| Question | Answer | Marks | Guidance |
|---|---|---|---|
| 3.i | $(c + is)^4 = c^4 + 4c^3(is) + 6c^2(-s^2) + 4c(is)^3 + (is)^4$ | **M1** | Uses binomial theorem to expand $(c + is)^4$. |
| | $\Rightarrow \cos4\theta = c^4 - 6c^2s^2 + s^4$ | **M1 A1** | Takes real part, AG. |
| ii | $\frac{\cos4\theta}{\cos^4\theta} = \tan^4\theta - 6\tan^2\theta + 1$ | **M1 A1** | Divides through by $\cos^4\theta$ |
| | Let $x = \tan\theta$, *then* $x^4 - 6x^2 + 1 = 0 \Rightarrow \cos4\theta = 0$<br>$\Rightarrow 4\theta = \pm\frac{\pi}{2} + 2m\pi,\ m \in \mathbb{Z}$ | **dM1** | Following on from finding correct quartic<br>Solves $\cos4\theta = 0$. |
| | Roots are $\tan q\pi$ where $q = \frac{1}{8}, \frac{3}{8}, \frac{5}{8}, \frac{7}{8}$.<br>$(q > 0)$ | **A1 A1** | **Alt methods:** Solves $\tan(4\theta) = (4t - t^3) / (t^4 - 6t^2 + 1)$ and $\tan(4\theta) = \infty$.<br>Solves equation in $\cot(\theta)$ after dividing by $\sin^4\theta$ |
| | | **8** | |

Explanation

3.i

$e^{4i\theta} = cos\,4\theta + isin\,4\theta = \left(e^{i\theta}\right)^4 = (cos\,\theta + isin\,\theta)^4$ **M1**

Consider real terms only:

$cos\,4\theta = cos^4\theta + 4C_2\,cos^2\theta(isin\theta)^2 + i^4\,sin^4\theta$

$\cos 4\theta = cos^4\theta - 6\,cos^2\theta\sin^2\theta + sin^4\theta$ **M1 A1**

ii)

let x=tanθ $0 < \theta < \pi\ (\theta \neq \frac{\pi}{2})$

then the equation becomes$tan^4\theta - 6\,tan^2\theta + 1 = 0$

$\cos 4\theta = cos^4\theta - 6\,cos^2\theta\sin^2\theta + sin^4\theta$

$\rightarrow \frac{\cos 4\theta}{\cos^4\theta} = 1 - 6\tan^2\theta + \tan^4\theta$ **M1 A1**

$cos\,4\theta = 0\ (0 < \theta < \pi)$

$0 < 4\theta < 4\pi$

$4\theta = \frac{\pi}{2},\ \frac{3}{2}\pi, \frac{5}{2}\pi, \frac{7}{2}\pi$ **dM1**

$\theta = \frac{\pi}{8},\ \frac{3}{8}\pi, \frac{5}{8}\pi, \frac{7}{8}\pi$

$x = tan\left(\frac{\pi}{8}\right), tan\left(\frac{3}{8}\pi\right), tan\left(\frac{5}{8}\pi\right), tan\left(\frac{7}{8}\pi\right)$**A1 A1**

Comment

i. Most candidates used de Moivre's Theorem as required to verify the expression for cos(4$\theta$), with a very small number wrongly attempting the verification using trigonometric identities alone.

ii. Whilst many candidates then correctly divided the expression from **(i)** by $\cos^4(\theta)$ to produce a quartic in tan($\theta$), others divided by $\sin^4(\theta)$ but failed to realise that this leads to an expression in cot($\theta$). Some failed to make a link with the first part as required. A variety of correct methods, some involving cot(4$\theta$), but most arising from cos(4$\theta$) = 0, were seen. Whilst most candidates made sure their answers were in the correct form, and were distinct, a small number gave repeated roots or failed to match the required form, losing marks.

*Answers*: **(ii)** tan $q\pi$ where $q = \frac{1}{8}, \frac{3}{8}, \frac{5}{8}, \frac{7}{8}$ or equivalent.

NO. 5

i. Use de Moivre's theorem to show that

$$\sin 8\theta = 8 \sin\theta \cos\theta(1 - 10 \sin^2\theta + 24 \sin^4\theta - 16 \sin^6\theta).$$ [6]

ii. Use the equation $\frac{\sin 8\theta}{\sin 2\theta} = 0$ to find the roots of

$$16x^6 - 24x^4 + 10x^2 - 1 = 0$$

in the form $\sin k\pi$, where $k$ is rational. [4]

A level FURTHER MATHEMATICS 2018/Winter/Paper1_1/Q7

Mark scheme

| Question | Answer | Marks | Guidance |
|---|---|---|---|
| 7.i | Write $c = \cos\theta$, $s = \sin\theta$.<br>$\cos 8\theta + i\sin 8\theta = (c + is)^8$ | **M1** | Uses binomial theorem. |
| | $\Rightarrow \sin 8\theta = 8c^7s - 56c^5s^3 + 56c^3s^5 - 8cs^7$ | **A1** | |
| | $= 8cs(c^6 - 7c^4s^2 + 7c^2s^4 - s^6)$ | **M1** | Factorises. |
| | $c^6 - 7c^4s^2 + 7c^2s^4 - s^6 = (1 - s^2)^3 - 7(1 - s^2)^2s^2 + 7(1 - s^2)s^4 - s^6$ | **M1** | Uses $c^2 = 1 - s^2$. |
| | $= (1 - 3s^2 + 3s^4 - s^6) - 7(s^2 - 2s^4 + s^6) + 7(s^4 - s^6) - s^6$<br>$= 1 - 10s^2 + 24s^4 - 16s^6$ | **A1** | |
| | $\Rightarrow \sin 8\theta = 8cs(1 - 10s^2 + 24s^4 - 16s^6)$ | **A1** | AG |
| | | **6** | |
| ii | $\sin 2\theta = 2cs \Rightarrow \dfrac{\sin 8\theta}{\sin 2\theta} = 4\left(1 - 10s^2 + 24s^4 - 16s^6\right)$ | **M1** | Uses $\sin 2\theta = 2cs$ to relate with equation in part (i) |
| | $\sin 8\theta = 0 \Rightarrow \theta = \dfrac{n\pi}{8}$ | **M1** | Solves $\sin 8\theta = 0$ |
| | $x = \sin\dfrac{\pi}{8}$ | **A1** | Gives one correct solution |
| | $\sin\dfrac{n\pi}{8},\ n = \pm 1, \pm 2 \pm 3 \text{ or } 1, 2, 3, 9, 10, 11$ | **A1** | Gives five other solutions. Allow different values of $k$ as long as six distinct solutions are found. sin0 and $\sin\dfrac{\pi}{2}$ must be excluded. |

Explanation

7.i

$(e^{i\theta})^8 = e^{i8\theta}$

$(\cos\theta + i\sin\theta)^8 = \cos 8\theta + i\sin 8\theta$ **M1**

Consider imaginary part only:

$8\cos^7\theta \cdot (i\sin\theta) + 8C_3\cos^5\theta(i\sin\theta)^3 + 8C_5\cos^3\theta(i\sin\theta)^5 +$

$8C_7\cos\theta(i\sin\theta)^7 = i\sin 8\theta$

$8\sin\theta\cos^7\theta - 56\sin^3\theta\cos^5\theta + 56\sin^5\theta\cos^3\theta - 8\sin^7\theta\cos\theta =$ **A1**

$\sin 8\theta$

Factorise, and use $\cos^2\theta = 1 - \sin^2\theta$

$8\sin\theta\cos\theta((1-\sin^2\theta)^3 - 7(1-\sin^2\theta)^2\sin^2\theta + 7(1 -$

$\sin^2\theta)\sin^4\theta - \sin^6\theta)) = \sin 8\theta$

**M1 M1**

$\sin 8\theta = 8\sin\theta\cos\theta(1 + (-3-7)\sin^2\theta + (3+7\times 2+7)\sin^4\theta +$ **A1**

$(-1-7-7-1)\sin^6\theta)$

$\sin 8\theta = 8\sin\theta\cos\theta(1 - 10\sin^2\theta + 24\sin^4\theta - 16\sin^6\theta)$ **A1**

ii)

from the result in i)

$\sin 8\theta = 4\sin 2\theta(1 - 10\sin^2\theta + 24\sin^4\theta - 16\sin^6\theta)$

$\frac{\sin 8\theta}{\sin 2\theta} = 4(1 - 10\sin^2\theta + 24\sin^4\theta - 16\sin^6\theta)$ **M1**

For the equation $16x^6 - 24x^4 + 10x^2 - 1 = 0$

Let x=sinθ $-\frac{\pi}{2} \le \theta \le \frac{\pi}{2}$ (restrict θ in a one-one interval)

$\frac{\sin 8\theta}{\sin 2\theta} = 4\times 0 = 0$

Find θ that sin8θ=0 and sin2θ≠0 **M1**

$-4\pi \leq 8\theta \leq 4\pi$

$8\theta = 0, \pm\pi, \pm 2\pi, \pm 3\pi, , \pm 4\pi$

$\theta = 0, \pm\frac{\pi}{8}, \pm\frac{\pi}{4}, \pm\frac{3\pi}{8}, \pm\frac{\pi}{2}$

Check sin2θ≠0 then

$\theta = \pm\frac{\pi}{8}, \pm\frac{\pi}{4}, \pm\frac{3\pi}{8}$ (hexic equations have 6 roots) **A1**

$x = sin\left(\pm\frac{\pi}{8}\right), sin\left(\pm\frac{\pi}{4}\right), sin\left(\pm\frac{3}{8}\pi\right)$ **A1**

Comment

i. This question was tackled well by most candidates with few errors, though some did not show all the necessary working. Many candidates multiplied out $(1 - s^2)^3$ in preference to using a binomial expansion and errors did sometimes creep in here.

ii. Although most candidates made the connection between the two parts, a significant number did not check to make sure that they wrote down six distinct roots, and that they eliminated any roots of sin(2$\theta$).

*Answers*: **(ii)** $\sin\frac{n\pi}{8}$, $n = \pm 1, \pm 2 \pm 3 or 1, 2, 3, 9, 10, 11$

NO. 6

i. Use de Moivre's theorem to prove that

$$\tan 4\theta = \frac{4\tan\theta - 4\tan^3\theta}{1 - 6\tan^2\theta + \tan^4\theta}. \quad [5]$$

ii. Hence find the solutions of the equation

$$t^4 - 4t^3 - 6t^2 + 4t + 1 = 0,$$

giving your answers in the form tan $k\pi$, where $k$ is a rational number. [5]

A level FURTHER MATHEMATICS 2017/Summer/Paper1_3/Q7

Mark scheme

| Question | Answer | M |
|---|---|---|
| 7.i | $(c + is)^4 = c^4 + 4c^3(is) + 6c^2(is)^2 + 4c(is)^3 + (is)^4$ | |
| | Equate real and imaginary parts | |
| | $\cos\theta = c^4 - 6c^2s^2 + s^4$ | |
| | $\sin\theta = 4c^3s - 4cs^3$ | |
| | $\tan 4\theta = \frac{\left(4c^3s - 4cs^3\right) \div c^4}{\left(c^4 - 6c^2s^2 + s^4\right) \div c^4} \left( = \frac{4\tan\theta - 4\tan^3\theta}{1 - 6\tan^2\theta + \tan^4\theta} \right)$ | |
| | | |
| ii | $\tan\theta = -1 \Rightarrow t^4 - 4t^3 - 6t^2 + 4t + 1 = 0$ | |
| | so $4\theta = \frac{3\pi}{4}\left(, \frac{7\pi}{4}, \frac{11\pi}{4}, \frac{15\pi}{4}\right)$ | |
| | $t = \tan\frac{3\pi}{16}, \tan\frac{7\pi}{16}, \tan\frac{11\pi}{16}, \tan\frac{15\pi}{16}.$ Allow $\left(\frac{k}{4} - \frac{1}{16}\right)\pi \; k = 0,1,2,3$ oe | A |
| | **Total:** | |

Explanation

7.i

$$e^{4i\theta} = \left(e^{i\theta}\right)^4$$

$$\cos 4\theta + i\sin 4\theta = (\cos\theta + i\sin\theta)^4$$

$\cos 4\theta + i \sin 4\theta = \cos^4\theta + 4\cos^3\theta(i\sin\theta) + 6\cos^2\theta(i\sin\theta)^2 +$

$4\cos\theta(i\sin\theta)^3 + (i\sin\theta)^4$ **B1**

Equate real and imaginary parts. **M1**

$cos\,4\theta = cos^4\theta - 6\,cos^2\theta\,sin^2\theta + sin^4\theta$ **A1**

$sin\,4\theta = 4\,sin\,\theta\,cos^3\theta - 4\,sin^3\theta\,cos\,\theta$ **A1**

$tan\,4\theta = \frac{sin\,4\theta}{cos\,4\theta} = \frac{4\,sin\,\theta\,cos^3\theta - 4\,sin^3\theta\,cos\,\theta}{cos^4\theta - 6\,cos^2\theta\,sin^2\theta + sin^4\theta}$

Divided by $cos^4\theta$

$= \frac{4tan\theta - 4\,tan^3\theta}{1 - 6\,tan^2\theta + tan^4\theta}$ **A1**

ii.

Since$tan\,4\theta = \frac{4tan\theta - 4\,tan^3\theta}{1 - 6\,tan^2\theta + tan^4\theta}$

If$tan\,4\theta = -1$, then

$-1 = \frac{4tan\theta - 4\,tan^3\theta}{1 - 6\,tan^2\theta + tan^4\theta}$

$-4tan\theta + 4\,tan^3\theta = 1 - 6\,tan^2\theta + tan^4\theta$

$tan^4\theta - 4\,tan^3\theta - 6\,tan^2\theta + 4tan\theta + 1 = 0$

**M1 M1**

$t = tan\,\theta\left(-\frac{\pi}{2} < \theta < \frac{\pi}{2}\right)$

The equation becomes$t^4 - 4t^3 - 6t^2 + 4t + 1 = 0$

$tan\,4\theta = -1 \rightarrow 4\theta = \frac{7}{4}\pi, \frac{3}{4}\pi, -\frac{1}{4}\pi, -\frac{5}{4}\pi$**M1**

$\theta = \frac{7}{16}\pi, \frac{3}{16}\pi, -\frac{1}{16}\pi, -\frac{5}{16}\pi$

$t = tan\,\theta = \tan\left(\frac{7}{16}\pi\right), \tan\left(\frac{3}{16}\pi\right), \tan\left(-\frac{1}{16}\pi\right), \tan\left(-\frac{5}{16}\pi\right)$

**A1 A1**

Comment

Most candidates were able to use de Moivre's Theorem and equate real and imaginary parts to find tan(4$\theta$) in **(i)**, though a number did not show sufficient working or did not deal correctly with the factor of i. A small number tried to use the expansion

of $(z + 1/z)^4$ and $(z – 1/z)^4$ but then encountered difficulties with substituting. In **(ii)** it was pleasing that many candidates related the necessary quartic equation to the trigonometric equation from **(i)**. Stronger candidates then solved the trigonometric equation and knew that four different answers were required to solve fully the quartic equation. These candidates usually gave their answers in the required form by either listing their solutions or stating the correct values for their parameter in a general solution. A small number of candidates forgot to specify values for the parameter.

*Answer*: **(ii)** $t = \tan\frac{3\pi}{16}, \tan\frac{7\pi}{16}, \tan\frac{11\pi}{16}, \tan\frac{15\pi}{16}$ or equivalent.

NO. 7

i. Use de Moivre's theorem to show that

$$\sin 5\theta = 5\sin\theta - 20\sin^3\theta + 16\sin^5\theta. \quad [5]$$

ii. Hence explain why the roots of the equation $16x^4 - 20x^2 + 5 = 0$ are $x = \pm\sin\frac{1}{5}\pi$ and $x = \pm\sin\frac{2}{5}\pi$. [3]

iii. Without using a calculator, find the exact values of

$$\sin\tfrac{1}{5}\pi\sin\tfrac{2}{5}\pi\sin\tfrac{3}{5}\pi\sin\tfrac{4}{5}\pi \quad \text{and} \quad \sin^2\left(\tfrac{1}{5}\pi\right) + \sin^2\left(\tfrac{2}{5}\pi\right). \quad [4]$$

A level FURTHER MATHEMATICS 2017/Winter/Paper1_1/Q10

Mark scheme

| Question | Answer | Marks | Guidar |
|---|---|---|---|
| 10.i | $\sin5\theta = Im(c + is)^5 = Im[c^5 + 5c^4is + 10c^3(is)^2 + 10c^2(is)^3 + 5c(is)^4 + (is)^5]$ | **B1** | SOI |
| | $\sin5\theta = 5c^4s - 10c^2s^3 + s^5$ | **M1A1** | |
| | $= s(5[1 - s^2]^2 - 10s^2[1 - s^2] + s^4)$ | **M1** | |
| | $= \ldots = 5\sin\theta - 20\sin^3\theta + 16\sin^5\theta$ | **A1** | AG |
| | | **5** | |
| ii | If $\theta = 0, \pm\frac{1}{5}\pi, \pm\frac{2}{5}\pi$ then $\sin 5\theta = 0$ | **B1** | |
| | $\Rightarrow 16s^5 - 20s^3 + 5s = 0$, where $s = \sin\theta$,<br>$\Rightarrow s(16s^4 - 20s^2 + 5) = 0$ | **B1** | |
| | $s = 0 \Rightarrow \theta = 0$ | **B1** | |
| | Hence roots of $16s^4 - 20s^3 + 5 = 0$ are $\pm\sin\frac{1}{5}\pi, \pm\sin\frac{2}{5}\pi$ | | AG |
| | | **3** | |
| iii | Since $\sin\frac{4}{5}\pi = -\sin\left(-\frac{1}{5}\pi\right)$ and $\sin\frac{3}{5}\pi = -\sin\left(-\frac{2}{5}\pi\right)$ | **B1** | |
| | $\sin\left(\frac{4}{5}\pi\right)\sin\left(\frac{3}{5}\pi\right)\sin\left(\frac{2}{5}\pi\right)\sin\left(\frac{1}{5}\pi\right) =$<br>$\sin\left(-\frac{1}{5}\pi\right)\sin\left(-\frac{2}{5}\pi\right)\sin\left(\frac{1}{5}\pi\right)\sin\left(\frac{2}{5}\pi\right) = \frac{5}{16}$ | **M1A1** | |
| | $\sin^2\frac{1}{5}\pi + \sin^2\frac{2}{5}\pi = -\frac{(-20)}{16} = \frac{5}{4}$ | **A1** | |
| | | **4** | |

Explanation

10.i

$e^{i5\theta} = \left(e^{i\theta}\right)^5$

$cos\,5\theta + i\,sin\,5\theta = (cos\,\theta + i\,sin\theta)^5$**B1**

Consider imaginary part only:

$i\,sin\,5\theta = 5C_1(cos\,\theta)^4 i\,sin\,\theta + 5C_3(cos\,\theta)^2(i\,sin\,\theta)^3 + (i\,sin\,\theta)^5$**M1A1**

$sin\,5\theta = 5\,sin\,\theta(1 - sin^2\,\theta)^2 - 10sin^3\,\theta(1 - sin^2\,\theta) + sin^5\,\theta$ **M1**

$sin\,5\theta = 5\,sin\,\theta(1 - 2\,sin^2\,\theta + sin^4\,\theta) - 10sin^3\,\theta(1 - sin^2\,\theta) +$

$sin^5\,\theta$

$sin\,5\theta = 5\,sin\,\theta - 20\,\sin^3\theta + 16\,\sin^5\theta$**A1**

10.ii

$sin\,5\theta = 5\,sin\,\theta - 20\,\sin^3\theta + 16\,\sin^5\theta = sin\,\theta\,(16sin^4\,\theta - 20\,\sin^2\theta + 5)$

Let x=sin θ

$\frac{sin\,5\theta}{sin\,\theta} = 16x^4 - 20x^2 + 5$

when$\frac{sin\,5\theta}{sin\,\theta} = 0, 16x^4 - 20x^2 + 5 = 0$

solve$\frac{sin\,5\theta}{sin\,\theta} = 0$:

$\theta = \pm\frac{\pi}{5},\ \pm\frac{2}{5}\pi$

Thus$x = sin\left(\pm\frac{\pi}{5}\right), sin\left(\pm\frac{2}{5}\pi\right) = \pm sin\left(\frac{\pi}{5}\right), \pm sin\left(\frac{2}{5}\pi\right)$

10.iii

$\pm sin\left(\frac{\pi}{5}\right), \pm sin\left(\frac{2}{5}\pi\right)$ are roots of$16x^4 - 20x^2 + 5 = 0$

Use the root properties of polynomials:

$\alpha\beta\gamma\delta = sin\frac{\pi}{5} \cdot \left(-sin\frac{\pi}{5}\right) \cdot sin\frac{2\pi}{5} \cdot \left(-sin\frac{2\pi}{5}\right) = \frac{5}{16}$

$\sum\alpha\beta = -sin^2\frac{\pi}{5} - sin^2\frac{2\pi}{5} = -\frac{20}{16} = -\frac{5}{4}$ (Totally 6 terms but 4 cancelled)

so$sin^2\frac{\pi}{5}sin^2\frac{2\pi}{5} = \frac{5}{16}$

$sin\frac{\pi}{5}sin\frac{2\pi}{5}sin\frac{3\pi}{5}sin\frac{4\pi}{5} = \frac{5}{16}\left(sin\frac{\pi}{5} = sin\frac{4\pi}{5}, sin\frac{2\pi}{5} = sin\frac{3\pi}{5}\right)$**B1 M1A1**

$sin^2\frac{\pi}{5} + sin^2\frac{2\pi}{5} = \frac{5}{4}$**A1**

Comment

i. Most candidates used de Moivre's theorem directly to find the given result, a few used $z - \frac{1}{z} 2isin(\theta)$ which took considerable manipulation. Although it was possible to obtain the result through using trigonometric relationships, this did not meet the criterion of the question.

ii. Following the instruction ‘hence’, the strongest candidates explained why the roots were the given pairs, either by using the related quadratic equation or by ensuring that the four roots they gave were distinct. They also justified the omission of $\sin(\theta) = 0$ as a solution. Some weaker solutions included errors such as $n\left(\frac{4\pi}{5}\right) = -\sin\left(\frac{\pi}{5}\right)$.

iii. Candidates using the related quadratic equation produced some good solutions to this part, others were able to relate $n\left(\frac{4\pi}{5}\right)$ to $-\sin-\left(\frac{\pi}{5}\right)$ correctly, but mistakes made above were repeated here leading some candidates to reach the correct answers from wrong working.

*Answers*: **(iii)** $\frac{5}{16}$ and $\frac{5}{4}$

NO. 8

a. Use de Moivre's theorem to show that $\cot 6\theta = \dfrac{\cot^6\theta - 15\cot^4\theta + 15\cot^2\theta - 1}{6\cot^5\theta - 20\cot^3\theta + 6\cot\theta}$. [6]

b. Hence obtain the roots of the equation

$$x^6 - 6x^5 - 15x^4 + 20x^3 + 15x^2 - 6x - 1 = 0$$

in the form $\cot(q\pi)$, where $q$ is a rational number. [4]

A level FURTHER MATHEMATICS 2024/Winter/Paper2_2/Q4

Mark scheme

| Question | Answer | Marks | Guida |
|---|---|---|---|
| a | $\cos 6\theta = \text{Re}(\cos\theta + i\sin\theta)^6 = \cos^6\theta - 15\cos^4\theta\sin^2\theta + 15\cos^2\theta\sin^4\theta - \sin^6\theta$ | M1A1 | Expand and tak real pa |
| | $\cos 6\theta = \text{Im}(\cos\theta + i\sin\theta)^6 = 6\cos^5\theta\sin\theta - 20\cos^3\theta\sin^3\theta + 6\cos\theta\sin^5\theta$ | M1A1 | Takes imagina part |
| | $\cot 6\theta = \dfrac{\cos^6\theta - 15\cos^4\theta\sin^2\theta + 15\cos^2\theta\sin^4\theta - \sin^6\theta}{6\cos^5\theta\sin\theta - 20\cos^3\theta\sin^3\theta + 6\cos\theta\sin^5\theta} \times \dfrac{\sin^{-6}\theta}{\sin^{-6}\theta}$ | M1 | Divides numera and denomi by $\sin^6$ |
| | $\cot 6\theta = \dfrac{\cot^6\theta - 15\cot^4\theta + 15\cot^2\theta - 1}{6\cot^5\theta - 20\cot^3\theta + 6\cot\theta}$ | A1 | AG |
| | | 6 | |
| b | $x = \cot\theta$, $\cot 6\theta = 1$ | B1 | Applies identity given ir |
| | $6\theta = \frac{1}{4}\pi + k\pi$ | M1 | Solves $\cot 6\theta =$ |
| | $\cot\left(\frac{1}{24}\pi\right)$ | A1 | Gives c correct solution 7.60 |
| | $\cot\left(\frac{5}{24}\pi\right), \cot\left(\frac{3}{8}\pi\right), \cot\left(\frac{13}{24}\pi\right), \cot\left(\frac{17}{24}\pi\right), \cot\left(\frac{7}{8}\pi\right)$ | A1 | Gives c five solution 1.30, 0 –0.13, 0.77, – Withho mark fc repeate roots. |
| | | 4 | |

Comment

a. After expanding $(\cos\theta + i\sin\theta)^6$ isin using the binomial expansion, most candidates accurately grouped together terms contributing to the real and

imaginary parts. Good responses then showed clear steps when dividing by $\sin^6\theta$ to fully justify the given result.

b. Almost all candidates rearranged the polynomial equation correctly, linking with the previous part of the question by substituting $x = \cot\theta$ . When solving $\cot 6\theta = 1$ , most candidates obtained $x = \cot\left(\frac{1}{24}\pi\right)$ . A few candidates gave the values for $\theta$ rather than actual solutions as instructed in the question.

NO. 9

a. Show that $\sum_{r=1}^{n} z^{4r} = \frac{z^{4n+2} - z^2}{z^2 - z^{-2}}$, for $z^2 \neq z^{-2}$. [2]

b. By letting $z = \cos\theta + i\sin\theta$, show that, if $\sin 2\theta \neq 0$,

$$\sum_{r=1}^{n} \sin(4r\theta) = \frac{\cos 2\theta - \cos(4n+2)\theta}{2\sin 2\theta}.$$ [5]

A level FURTHER MATHEMATICS 2024/Summer/Paper2_3/Q6

Mark scheme

| Question | Answer | Marks | Guidance |
|---|---|---|---|
| a | $z^4 + z^4(z^4) + \ldots + z^4(z^4)^{n-1} = \frac{z^4\left(z^{4n}-1\right)}{z^4-1}$ | M1 | Uses sum o geometric series. |
| | $\frac{z^4\left(z^{4n}-1\right)}{z^4-1} \times \frac{z^{-2}}{z^{-2}} = \frac{z^{4n+2}-z^2}{z^2-z^{-2}}$ | A1 | Divides numerator and denominato by $z^2$, AG. |
| | | 2 | |
| b | $z^2 - z^{-2} = 2\text{i}\sin 2\theta$ | B1 | Simplifies denominato |
| | $\frac{z^{4n+2}-z^2}{z^2-z^{-2}} = \frac{\cos(4n+2)\theta + \text{i}\sin(4n+2)\theta - \cos 2\theta - \text{i}\sin 2\theta}{2\text{i}\sin 2\theta}$ | M1 A1 | Applies de Moivre's theorem to numerator. (Used at least once f M1.) Accep $z^2 = \cos^2\theta +$ $2\text{i}\sin\theta\cos\theta -$ $\sin^2\theta$. |
| | $\sum_{r=1}^{n} \sin\left(4r\theta\right) = \frac{\cos(4n+2)\theta - \cos 2\theta}{-2\sin 2\theta}$ | M1 | Equates imaginary parts. |
| | $\sum_{r=1}^{n} \sin\left(4r\theta\right) = \frac{\cos 2\theta - \cos(4n+2)\theta}{2\sin 2\theta}$ | A1 | AG. Must s evidence of eliminating from the denominato |
| | | 5 | |

Comment

a. The majority of candidates used the formula for the sum of a geometric progression correctly. Good candidates showed their working clearly when dividing the numerator and denominator by $z^2$ to fully justify the given answer.

b. Almost all knew that de Moivre's theorem related the series to the geometric progression in part **(a)**. Strong candidates accurately took the imaginary part, after simplifying the numerator and denominator, which led to the given answer.

Often, crucial steps were omitted, and therefore answers were not full and detailed enough.

NO. 10

By considering the binomial expansions of $\left(z+\frac{1}{z}\right)^4$ and $\left(z-\frac{1}{z}\right)^4$, where $z = \cos\theta + i\sin\theta$, use de Moivre's theorem to show that

$$\cot^4\theta = \frac{\cos 4\theta + a\cos 2\theta + b}{\cos 4\theta - a\cos 2\theta + b},$$

where $a$ and $b$ are integers to be determined. [7]

A level FURTHER MATHEMATICS 2023/Summer/Paper2_3/Q3

Mark scheme

| Question | Answer | Partial Marks | Guidance |
|---|---|---|---|
| 3 | $z - z^{-1} = 2i\sin\theta$ and $z + z^{-1} = 2\cos\theta$ | B1 | Use of $z - z^{-1} = 2i\sin\theta$ and $z + z^{-1} = 2\cos\theta$ |
| | $(z + z^{-1})^4 = (z^4 + z^{-4}) + 4(z^2 + z^{-2}) + 6$ | M1 A1 | Expands and groups. |
| | $(z - z^{-1})^4 = (z^4 + z^{-4}) - 4(z^2 + z^{-2}) + 6$ | M1 A1 | Expands and groups. Only withhold one A1 mark (this or the previous) for no clear grouping. |
| | $\frac{2^4\cos^4\theta}{2^4\sin^4\theta} = \frac{2\cos 4\theta + 4(2\cos 2\theta) + 6}{2\cos 4\theta - 4(2\cos 2\theta) + 6}$ | M1 | Substitutes $z^n + z^{-n} = 2\cos n\theta$ once in LHS. |
| | $\cot^4\theta = \frac{\cos 4\theta + 4\cos 2\theta + 3}{\cos 4\theta - 4\cos 2\theta + 3}$ | A1 | |
| | | 7 | |

Comment

After expanding $(z + z^{-1})^4$ and $(z - z^{-1})^4$ using the binomial expansion, good candidates grouped together terms clearly before applying the identity $z^n + z^{-n} = 2\cos n\theta$ to fully justify their answer. Similarly, for the lefthand side, evidence was needed that candidates had used both $z - z^{-1} = 2i\sin\theta$ and $z + z^{-1} = 2\cos\theta$; it was sometimes not apparent that they knew these rules.

NO. 11

a. State the sum of the series $1 + w + w^2 + w^3 + ... + w^{n-1}$, for $w \neq 1$. [1]

b. Show that $(1 + i\tan\theta)^k = \sec^k\theta(\cos k\theta + i\sin k\theta)$, where $\theta$ is not an integer multiple of $\frac{1}{2}\pi$. [2]

c. By considering $\sum_{k=0}^{n-1}(1+i\tan\theta)^k$, show that

$$\sum_{k=0}^{n-1}\sec^k\theta\sin k\theta = \cot\theta(1-\sec^n\theta\cos n\theta),$$

provided $\theta$ is not an integer multiple of $\frac{1}{2}\pi$. [5]

d. Hence find $\sum_{k=0}^{6m-1}2^k\sin\left(\frac{1}{3}k\pi\right)$ in terms of $m$. [2]

A level FURTHER MATHEMATICS 2022/Winter/Paper2_1/Q7

Mark scheme

| Question | Answer | Marks | Gu |
|---|---|---|---|
| a | $\frac{w^n-1}{w-1}$ | B1 | |
| | | 1 | |
| b | $(1+i\tan\theta)^k = \sec^k\theta(\cos\theta + i\sin\theta)^k = \sec^k\theta(\cos k\theta + i\sin k\theta)$ | M1 A1 | App Moi thec AG. |
| | | 2 | |
| c | $\sum_{k=0}^{n-1}(1+i\tan\theta)^k = \frac{(1+i\tan\theta)^n-1}{i\tan\theta}$ | M1 A1 | App **(a)**. |
| | $-\cot\theta\sec^n\theta(i\cos n\theta - \sin n\theta) + i\cot\theta$ | A1 | |
| | $\sum_{k=0}^{n-1}\sec^k\theta\sin k\theta = \cot\theta\left(1-\sec^n\theta\cos n\theta\right)$ | M1 A1 | Take ima part |
| | | 5 | |
| d | $\theta = \frac{1}{3}\pi \quad \sum_{k=0}^{6m-1}2^k\sin\left(\frac{1}{3}k\pi\right) = \frac{1}{3}\sqrt{3}\left(1-2^{6m}\cos\left(2m\pi\right)\right) = \frac{1}{3}\sqrt{3}\left(1-2^{6m}\right)$ | M1 A1 | Sets |
| | | 2 | |

Comment

a. The majority of candidates used the formula for the sum of a geometric progression accurately.

b. The majority of candidates applied de Moivre's theorem to fully justify the given result.

c. Almost all candidates knew that de Moivre's theorem related the series to the geometric progression in **(a)**. Stronger candidates accurately took the imaginary part, after simplifying the numerator and denominator, which led to the given answer.

d. Stronger candidates maintained accuracy when substituting $\theta = \frac{1}{3}\pi$ and $n$ = 6m, fully simplifying their answer.

NO. 12

a. Find $a$ and $b$ such that $z^8 - iz^5 - z^3 + i = (z^5 - a)(z^3 - b)$. [1]

b. Hence find the roots of

$$z^8 - iz^5 - z^3 + i = 0,$$

giving your answers in the form $re^{i\theta}$, where $r > 0$ and $0 \leqslant \theta < 2\pi$. [6]

A level FURTHER MATHEMATICS 2021/Summer/Paper2_3/Q1

Mark scheme

| Question | Answer | Marks | Guidance |
|---|---|---|---|
| a | $a = 1$, $b = i$ | **B1** | |
| | | **[1]** | |
| b | $z = e^{i\frac{2k}{5}\pi}$ | **M1** | Finds one fifth root of unity. |
| | $z = e^{i\frac{2k}{5}\pi}$, $k = 0,1,2,3,4$ | **A1** | Gives all fifth roots of unity. Accept 1 not in exponential form. A0 if $r = 1$ not seen or implied. |
| | Argument of i is $\frac{1}{2}\pi$. | **B1** | |
| | $z = e^{i\frac{1}{6}\pi}$ | **M1** **A1** | Finds one root of $z^3 = i$. A0 if first root not given in exponential form. |
| | $z = e^{i\frac{5}{6}\pi}$, $e^{i\frac{3}{2}\pi}$ | **A1** | Finds other two roots of $z^3 = i$. A0 if $r = 1$ not seen or implied. Withhold final A1 if $a$ and $b$ reversed in part **(b)** but all other work correct. |
| | | [6] | **SC** If $z = e^{i\left(\frac{1}{10}\pi + \frac{2k}{5}\pi\right)}$, $k = 0,1,2,3,4$ award M1 A1 A1 only. **SC** If $z = e^{i\left(\frac{2k}{3}\pi\right)}$, $k = 0,1,2$ award M1 A1 only. |

Explanation

a)

RHS $= z^8 - bz^5 - az^3 + ab$=LHS

Compare the coefficients:

b=i, a=1 **B1**

b)

$z^8 - iz^5 - z^3 + i = (z^5 - 1)(z^3 - i) = 0$

$z^5 = cos\,5\theta + i\,sin\,5\theta = 1, or\ z^3 = \cos 3\theta + i \sin 3\theta = i$ **B1**

$0 \le \theta < 2\pi$, so $0 \le 5\theta < 10\pi, 0 \le 3\theta < 6\pi$

$cos\,5\theta = 1,\ sin\,3\theta = 1$ **M1 M1**

$5\theta = 0, 2\pi, 4\pi, 6\pi, 8\pi\ \ or\ \ 3\theta = \frac{\pi}{2}, \frac{5\pi}{2}, \frac{9\pi}{2}$

$\theta = 0, \frac{2\pi}{5}, \frac{4\pi}{5}, \frac{6\pi}{5}, \frac{8\pi}{5}, \frac{\pi}{6}, \frac{5\pi}{6}, \frac{9\pi}{6}$

$z = 1, e^{\frac{2}{5}\pi i}, e^{\frac{4}{5}\pi i}, e^{\frac{6}{5}\pi i}, e^{\frac{8}{5}\pi i}, e^{\frac{1}{6}\pi i}, e^{\frac{5}{6}\pi i}, e^{\frac{3}{2}\pi i}$ **A1 A1 A1**

Comment

a. Almost all candidates stated the correct values of *a* and *b*.

b. Most candidates found the fifth roots of 1 and the cube roots of i. Some candidates reversed the values of *a* and *b*. Better responses showed clear working, starting from 1 and i in exponential form, and listed all eight roots in the required form.